TEXORA · LECTURE SHEET

Washing: Machineries Used for Garments Washing

The garment wash plant is a small factory of its own: washers sized by the load, extraction and drying that must keep pace, and effect machines — sand blasting cabinets, PP spray booths, laser and ozone — arranged around the recipe of Topic 5.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 10 · SPECIALIZATION — WET PROCESSING

1. The wash plant as a production line

Garment washing processes made-up garments in loads, and its machinery must be balanced as a line: washers create the effect, hydro-extractors remove water, tumble dryers set the final appearance, and effect stations (hand sanding, whiskering, PP spray, tagging) work on dry garments before or between wet steps. A bottleneck anywhere — usually drying — limits the whole plant, so capacities are matched in loads per shift, not in machine counts.

2. Washing machines

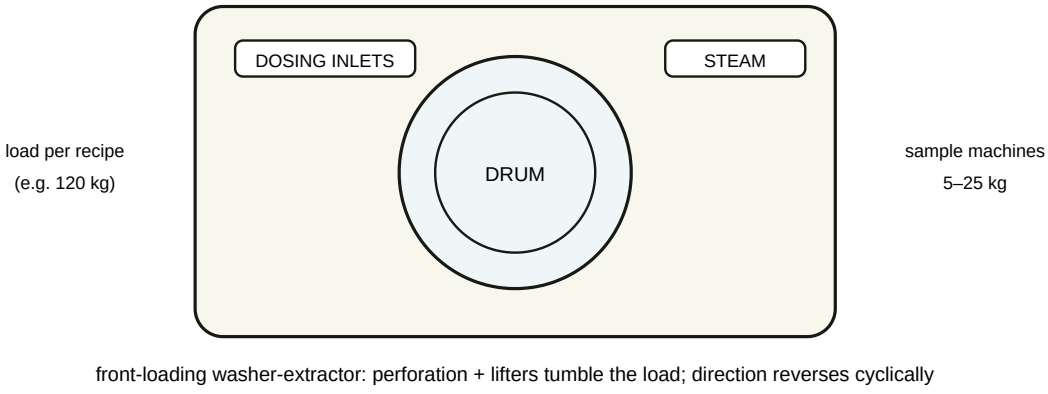


Fig. 1 — Front-loading garment washing machine (schematic): a perforated horizontal drum with lifters; load ratio, drum speed and reversal rhythm supply the mechanical action the recipe assumes.

- **Front-loading washer-extractors:** the production standard — horizontal perforated drum, programmable fill, temperature, drum speed and reversal; extraction in the same drum at high speed. Capacities from 25 kg samples machines to several hundred kilograms.

- **Side-loading (Y-pocket) washers:** drum divided into compartments; even gentler and easier loading of bulky denim loads.
- **Dyeing/washing drums for garments:** paddle and rotary drum machines also used for garment dyeing (WP-4101 Topic 3).
- **Sample washers:** small exact-scale machines (5–25 kg) on which wash recipes are developed so that load ratio and mechanical action reproduce production — a recipe developed in a bucket does not transfer.

Mechanical action is a machine variable the recipe depends on: drum diameter, lifter design, speed and reversal rhythm set how hard garments are tumbled and abraded. Two machines at the same load and recipe but different action give different washes — one reason recipes are proved on the production machine class (Topic 5).

3. Water removal and drying

- **Hydro-extractor:** a vertical centrifuge basket spinning the load to remove most bath water in minutes; reduces dryer load dramatically. Garments are loaded evenly to avoid violent imbalance.
- **Tumble dryer:** heated air through a rotating perforated drum; the final appearance (softness, shrinkage completion, indigo rub-off) is set here. Overloading or over-drying shows as streaks, excessive creasing and strength loss; temperature is limited for elastane-containing garments.
- **Steam tunnel / press finishing:** for washed garments needing a formed finish before packing.

WORKED EXAMPLE 1 — LINE BALANCING IN LOADS PER SHIFT

Washer-extractor: 100 kg per load, wash cycle 45 min → in an 8 h shift: ⌊480/45⌋ = 10 loads = **1,000 kg** of garments per machine per shift.
Hydro-extraction is fast (about 10 min per load) and never limits. The tumble dryer takes 50 kg per load and 40 min per cycle: per shift ⌊480/40⌋ = 12 loads × 50 kg = 600 kg — **the dryer limits the line**. To keep pace with one washer (1,000 kg), the plant needs ⌈1000/600⌉ = **2 dryers** of this size (or one larger/faster unit).
Line balancing in loads, cycle times and the floor function — the same arithmetic as Topic 7 — is how wash plants are actually sized.

4. Effect machinery (dry-process stations)

STATION	FUNCTION
Hand sanding / whiskering boards and	Local abrasion on inflated garments to raise whiskers and worn areas before washing

inflatable forms	
PP spray booth	Potassium permanganate sprayed on local areas (Topic 5), with extraction and operator protection
Laser engraving machine	CO_2 laser burns surface indigo to a programmed pattern — whiskers, chevrons, all-over designs; repeatability is digital
Ozone machine	Gas-phase bleaching/lightening of dry or damp garments, replacing much hypochlorite work
Tagging / grinding stations	Localised edge abrasion and distressing with small tools
Curing oven (for resin whiskers)	3D whisker shapes are resin-set and cured before washing so the creases survive the wash

5. Utilities and effluent of the wash plant

Washing is water-intensive per garment, and denim effluent carries indigo, pumice solids and (in bleach work) residual oxidant. Plants therefore include stone separation and screening at drain, and schedule loads to smooth water demand. Laser and ozone stations (Topic 11) are adopted largely to cut this water and chemical load per garment while adding digital repeatability — an effects decision that is also an effluent decision.

6. Glossary — key terms

- **Washer-extractor:** a machine combining washing and centrifugal extraction in one drum.
- **Hydro-extractor:** a centrifuge removing water from washed garments by spin.
- **Load ratio:** bath volume per mass of garments in a wash load (the wash-house LR).
- **Dry process:** effects produced on dry garments (sanding, spraying, laser) before or between wet processes.

7. Key points

- The wash plant is balanced in loads per shift; dryers, not washers, usually limit output.
- Mechanical action is part of the machine's identity — recipes transfer only between machines of similar action, proved on sample washers of the same class.
- Effect stations are production machinery, scheduled and costed like washers, and increasingly chosen for water and chemical savings.

8. Summary

Garment washing machinery turns the recipes of Topic 5 into loads per shift: washer-extractors sized to the load, extraction and drying balanced against them (two 50 kg dryers behind one 100 kg washer in the worked example), and a dry-process line whose tools — hand, spray, laser, ozone — decide the effects the wash will reveal.

9. Practice

1. A washer takes 150 kg per load with a 40 min cycle. A dryer takes 75 kg per load with a 35 min cycle. In an 8 h shift, how many dryers are needed to balance one washer?
2. Why are wash recipes developed on a 10–25 kg sample washer rather than in laboratory glassware?
3. Give two production advantages of laser over hand sanding for whisker effects, and one limitation.

Answer key

1. Washer: ⌊480/40⌋ = 12 loads × 150 kg = 1,800 kg. Dryer: ⌊480/35⌋ = 13 loads × 75 kg = 975 kg. Dryers needed = ⌈1800/975⌉ = 2.
2. Mechanical action — tumbling, lifter drop, load compaction — is a primary washing variable that glassware cannot reproduce. A sample washer of the same class scales the action with the load, so time/enzyme/stone predictions transfer to production.
3. Advantages: digital repeatability (every garment identical to the design file) and speed/no consumable abrasives (also removes silica-dust exposure of hand sanding with sandpaper and the banned sandblasting route). Limitation: capital cost and the need for careful power/focus calibration — wrong settings scorch or cut the fabric, and 3D hand-feel effects still need complementary manual work.
